

Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary –

Rows: 3,900

Columns: 18 –

Key Features: - Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● **Data Loading:** Imported the dataset using pandas.

● **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

● **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● **Column Standardization:** Renamed columns to snake case for better readability and documentation.

● **Feature Engineering:**

○ Created age_group column by binning customer ages.

○ Created purchase_frequency_days column from purchase data.

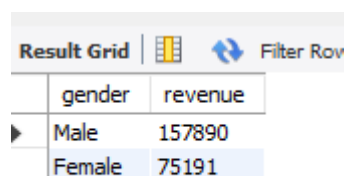
● **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4.Data Analysis using SQL (Business Transactions)

We performed structured analysis in MYSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'gender' and 'revenue'. The 'Male' row has a value of 157890, and the 'Female' row has a value of 75191. The 'Female' row is highlighted in blue.

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Result Grid			
		Filter Rows:	
		Export:	
		Wrap Cell Content:	
	customer_id	discount_applied	purchase_amount
▶	2	Yes	64
	3	Yes	73
	4	Yes	90
	7	Yes	85
	9	Yes	97
	12	Yes	68
	13	Yes	72
	16	Yes	81
	20	Yes	90
	22	Yes	62
	24	Yes	88
	29	Yes	94
	32	Yes	79

customer 3 x

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	round(avg(review_rating),2)
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

Result Grid		
		Filter Rows:
		Export:
		Wrap Cell Content:
	shipping_type	avg(purchase_amount)
▶	Express	60.4752
	Standard	58.4602

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Result Grid				
		Filter Rows:		
		Export:		
		Wrap Cell Content:		
	subscription_status	round(avg(purchase_amount),2)	sum(purchase_amount)	count(customer_id)
▶	Yes	59.49	62645	1053
	No	59.87	170436	2847

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	item_purchased	discount_rate				
▶	Hat	50.0000				
	Sneakers	49.6552				
	Coat	49.0683				
	Sweater	48.1707				
	Pants	47.3684				

- 7. Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	customer_segment	number_of_customer			
▶	LOYAL	3116			
	RETURNING	701			
	NEW	83			

- 8. Top 3 Products per Category** – Listed the most purchased products within each category.

Result Grid			Filter Rows:	Export:	Wrap Cell Conts
	category	item_purchased	total_order	item_rank	
▶	Accessories	Jewelry	171	1	
	Accessories	Sunglasses	161	2	
	Accessories	Belt	161	3	
	Clothing	Blouse	171	1	
	Clothing	Pants	171	2	
	Clothing	Shirt	169	3	
	Footwear	Sandals	160	1	
	Footwear	Shoes	150	2	
	Footwear	Sneakers	145	3	
	Outerwear	Jacket	163	1	
	Outerwear	Coat	161	2	

- 9. Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

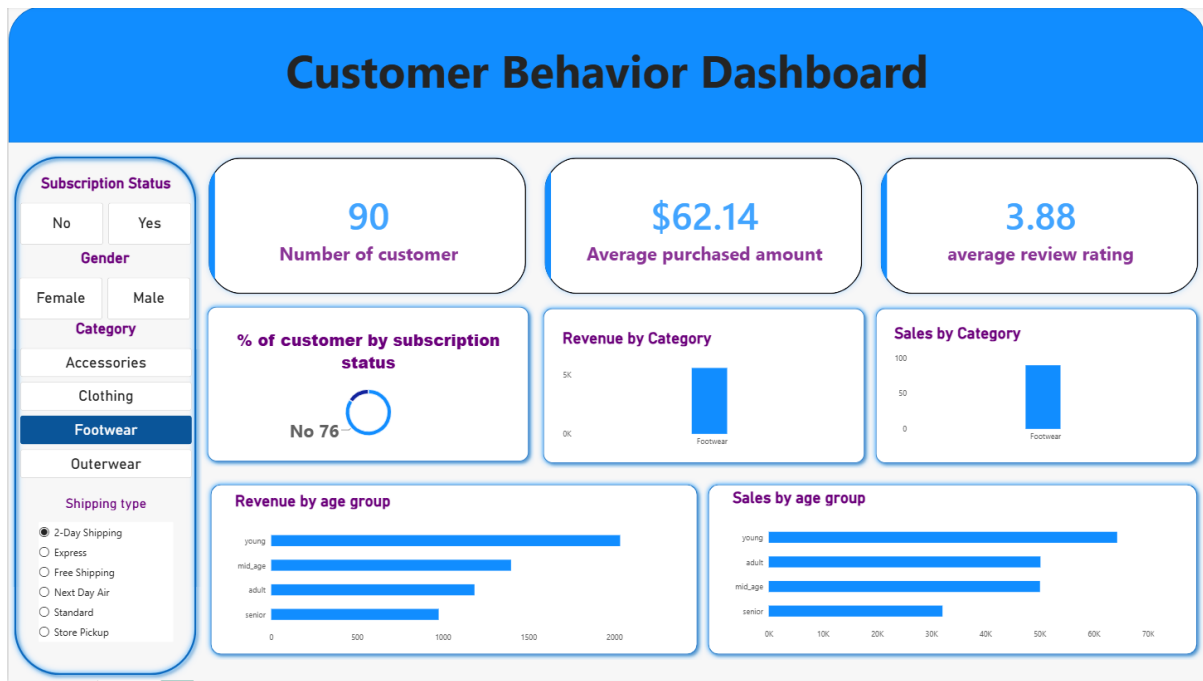
Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	subscription_status	repeat_buyers			
▶	Yes	958			
	No	2518			

- 10. Revenue by Age Group** – Calculated total revenue contribution of each age group.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	age_group	total_revenue	
	young	62143	
	mid_age	59197	
	adult	55978	
	senior	55763	

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.

- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.